

Neeraj Sharma

Principal Platform Engineer | AI Security Engineer · DSPM Architect · MLSecOps · DevSecOps

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SUMMARY

Principal-level engineer bridging 25+ years of large-scale infrastructure reliability with modern AI/ML security governance and GenAI platform engineering. Specialized in Zero Trust AI infrastructure, DSPM, and DevSecOps at scale — from IoT and global CDN platforms to cloud-native Kubernetes environments across IBM's 22+ global production data centers.

CORE SKILLS

DSPM | AI Security | AI Governance | GenAI | LLMs | Agentic AI | RAG | Vector Databases | LangChain/LangGraph | Python | PostgreSQL/SQL | Kubernetes | Terraform | Tekton CI/CD | IBM watsonx | IBM Cloud | GCP | Observability | Zero Trust | FedRAMP

Professional Development – Applied AI, GenAI & Cloud Security April 2025 - Present

Dedicated full-time to advancing expertise in AI, GenAI, and cloud security through postgraduate study, industry certifications, and production-oriented engineering projects.

- Developed an Agentic AI application for CVE + EPSS prioritization using Python, LangChain, LangGraph, and LangSmith (code available upon request).
- Designed and implemented an open-source, production-oriented multi-backend RAG system with enterprise retrieval architecture supporting a default local inference backend and interchangeable IBM watsonx and AWS Bedrock integrations.
- Applied DSPM and secure AI/ML pipeline principles to AI solution architectures.

WORK EXPERIENCE

IBM — San Jose, CA — Senior Backend Engineer (DevSecOps / SRE) Apr 2021 – Apr 2025

- Evolved the backend reliability and observability stack for IBM's global network underlay infrastructure spanning 22+ production data centers worldwide, owning service-level stability across 60+ microservices.
- Hardened release pipelines using Tekton CI/CD with integrated vulnerability management and patching; authored 200+ Public Cloud Exceptions (PCEs) and led cross-functional efforts with Dev, Security, and SRE teams to reduce the PCE backlog from 200+ to 30 — an 85% reduction in cloud security debt.
- Engineered and maintained 200+ Python and Ansible automation tests, eliminating ~40 hrs/month of repeated manual regression effort and improving deployment confidence at scale.
- Operated Kafka and Zookeeper within a K8s-based observability platform to troubleshoot production incidents across 60+ microservices; leveraged ELK stack and Prometheus/Grafana to isolate root causes and maintain 99.9% alert delivery reliability. Earned IBM Credly certifications including watsonx Essentials (watsonx.ai · Foundation Models · Generative AI), Cloud DevSecOps, Containers & Kubernetes, Trustworthy AI & AI Ethics, and Security & Privacy by Design; coursework coverage across watsonx.data and watsonx.governance via IBM Generative AI Engineering.

Exabeam — San Mateo, CA — Senior Software Quality Engineer (Security Analytics) Aug 2019 – Mar 2021

- Led validation engineering for a SIEM/UEBA platform processing billions of behavioral events daily across GCP, ensuring detection accuracy at scale for Advanced Analytics and correlated threat workflows.
- Augmented existing test frameworks with targeted event correlation tests — a key feature for filtering false positives — improving detection accuracy by 20% across distributed security analytics pipelines.
- Expanded data-driven security model coverage, reducing false-negative risk in production threat detection.

Oracle (OCI) — San Jose, CA — Principal Software Engineer May 2018 – Jul 2019

- Architected and validated multi-tenant leaf-spine network infrastructure across 20+ global OCI data centers; delivered automation microservices that accelerated new cloud region provisioning from weeks to days.

Light.co — Redwood City, CA — Senior Software Quality Engineer Jun 2017 – Apr 2018

- Built end-to-end test automation for the L16 computational camera system (16-lens array), validating firmware-to-Android integration across image capture, merge, and computational photography processing pipelines.

Silver Spring Networks — Redwood City, CA — Senior SDET Nov 2012 – May 2017

- Built E2E test automation from scratch for an IoT smart grid platform supporting 23M+ deployed devices across utility networks; achieved ~60% automation coverage within a 2-week firmware release cadence, significantly reducing manual validation risk at fleet scale.

Redwood Systems — Fremont, CA — Senior SDET Jan 2011 – Oct 2012

- Led system integration testing for sensor-driven smart lighting platforms deployed in enterprise data centers including Meta and Microsoft facilities, validating energy management and sensor control reliability.

CDNetworks — San Jose, CA — QA Manager Aug 2009 – Dec 2010

- Led QA strategy for a globally distributed CDN; established canary deployment validation pipelines that reduced production rollback risk and strengthened performance and data integrity across large-scale edge networks.

EDUCATION

PGP Artificial Intelligence & Deep Learning (AIDL) — UT Austin McCombs 2026

PGP Data Science & Business Analytics (PGPDSBA) — UT Austin McCombs 2025

Savitribai Phule Pune University — M.S. Computer Management 1995

SELECTED AI PLATFORM & DSPM PROJECTS

Full portfolio: github.com/neeraj-sharma-0

- DSPM – GenAI VectorDB Governance: data asset classification, column-level sensitivity detection, audit scoring, policy enforcement, and count-based deletion verification across Chroma, Pinecone, Qdrant, Weaviate, Milvus, and OpenSearch.
- DSPM – DevSecOps Ephemeral Compute Evidence Pipeline: DSPM controls for CI/CD to detect, audit, and manage sensitive data exposure in logs, artifacts, and ephemeral environments.